DECENTRALIZED FEDERATED LEARNING OVER IMPERFECT CHANNELS

Aditya Kumar Sahu, Rishav Kumar Manjhi, D. YASWANTH REDDY

Department of Computer Science and Engineering, Vellore Institute of Technology (VIT), Vellore, 632014, Tamil Nadu, India

Corresponding author: aditya.kumarsahu2024@vitstudent.ac.in

Abstract:

In federated learning (FL), devices collaborate to train a shared model while keeping their local data secure. But the wireless network used in the process is prone to packet losses and delays.

Few works investigate how network structure and the channel affect convergence rate and communication cost in decentralised FL. This work examines gossip communication in decentralised FL. It assesses the performance of the framework via three types of analyses.

Training of a convolutional neural network (CNN) model on the MNIST dataset is considered where five clients are interconnected with each other using Ring, Grid, and Erdős–Rényi random topologies. Packet loss ratios considered in the simulations are 0% and 20% while the simulation duration per experiment is set at 20 gossip rounds.

In all the three topologies, almost the same test accuracy of 99.1% is achieved. Even if there is a 20% loss ratio of packets. In addition to that, the consensus error in the case of Ring topology is 0.121 which is about 2.6 times less than that of the Grid topology which is 0.315.

Gossip communications in events have been utilized in this project. The clients will send information only when there is a change in the model above a certain level. This method decreases communication by up to 8%. The decrease in accuracy is not more than 0.03%.

The spectral gap analysis assists in explaining the rate of convergence. The larger the spectral gap, the faster the process of consensus. These results prove that decentralized FL is resilient to the loss of packets. They also provide ways to minimize communication in wireless edge networks.

Introduction

1.1 Background

The increased use of Internet-of-Things and Edge Computing has resulted in various opportunities for

collaborative machine learning at the network edge. Federated Learning, referred to as FL, as developed by McMahan et al., allows multiple clients to train a common machine learning model, without bringing data to a central location, and thus ensures user privacy. The typical Federated Learning framework relies on the existence of a central parameter server that collects updates from all participating clients. This architecture is very fragile in ad-hoc, vehicular and disaster response scenarios where there is not an existing infrastructure. In such scenarios, the clients must communicate through peer-to-peer connections with possible packet losses and variable delays.

1.2 State-of-the-art

Decentralized federated learning eliminates the use of server by gossiping of neighbors.

Every device trains its model and communicates with the nearby devices through sharing its model updates. They use a doubly stochastic matrix that averages out the updates where both rows and columns add up to 1[2].

Lian et al. [3] proved that for decentralized gradient descent with doubly stochastic matrix W the speed of convergence to steady-state is comparable with the centralized SGD provided that communication between

the devices is reliable. The rate of convergence is defined by spectral gap of W which is equal to $\gamma = 1. |\Lambda_2(W)|$, where λ_2 is the second-largest eigenvalue of W [4]. Greater spectral gap is better.

Event-triggered communication mechanism [5] decreases the amount of communications by transmitting information only when there were sufficient changes in the model since the last transmission.

1.3 The gap

Notwithstanding this literature review, no comprehensive research exists that investigates simultaneously the following: (i) the impact of channel characteristics like packet loss and latency on gossip-based FL performance in different types of network topology, (ii) the way of saving communication by transmitting information only when needed, and (iii) evaluating the time required for convergence to a specific consensus state. All of this in one repeatable experiment. Previous research either ignores the presence of communication channel imperfections or considers only one network structure, which makes it difficult to say whether it is the effect of network topology or channel quality.

1.4 Objectives

This research implements and evaluates the performance of gossip-based decentralized FL over wireless channels. In particular, it analyses the impact of network topology, packet loss rate, and event-triggered communication on model accuracy, consensus error, and communication cost.

1.5 Contributions

The main contributions in this research work are:

1. M1 – Gossip FL Simulator with user-defined topology (Ring, Grid, random),

Loss rate of packets, asynchronous delays, using Metropolis–Hastings mixing weights with

automatic normalization of lost packets.

2. M2 – Gossip protocol with an event-trigger mechanism that stops sending packets if ‖ $wi(t)$

− wi last ‖2≤ δ, analyzing the trade-off between accuracy and communication for different δ.

3. M3 – Framework to derive theoretical bounds on convergence rates (SLEM, mixing time) by

analyzing the spectral gap and empirical validation through gossip-based scalar simulation.

Literature review:

2.1 Decentralized Federated Learning & Gossip Based Optimization

Decentralized Federated Learning refers to several devices collaborating to learn the same model without any use of server. Decentralized Federated Learning does not communicate any information of updates of parameters to any coordinators. Data exchange happens among the neighboring nodes. Update the local model through Decentralized aggregation. Decentralized Federated Learning shares the responsibility of learning and also lowers the dependency on the point of coordination. Lian et al. [3] Have researched decentralized parallel gradient descent techniques, making comparison between decentralized optimization and centralized optimization, revealing the importance of Decentralized Federated Learning techniques. Nedic et al. [4] Have researched the impact of network topology on communication and computation in an optimization process.

The gossip algorithms serve to offer ways through which Decentralized Federated Learning may communicate. Koloskova et al.[5] analyzed Decentralized Federated Learning through the use of gossip algorithms and proved that reduction of communication was important in decentralized learning. Hegedűs et al. [6] studied gossip learning as a Decentralized Federated Learning alternative to lay the grounds for learning among peers without coordination. In the same spirit, Savazzi et al. [7] studied a consensus mechanism for Decentralized Federated Learning on largescale IoT networks. All these studies prove that decentralized aggregation and neighbor-to-neighbor communication is fundamental for Decentralized Federated Learning. However, the diverse algorithmic approaches used indicate the need for the testing of Decentralized Federated Learning under various conditions of network and communication.

2.2 Efficient Communication and Learning Using Inefficient Communication Means

In the case of increased model exchange, which leads to increased traffic, delay, and consumption of extra resources, efficient communication becomes one of the key components in the process of decentralized learning. To solve this problem, scientists dived into researching communications which deliver compressed information. The first concept is communication in decentralized stochastic optimization investigated by Koloskova et al., while the second is consensus-based learning in IoT networks investigated by Savazzi et al.

Wireless and distributed communication networks might be unreliable, slow, noisy, and prone to packet loss. This could stop devices from receiving the models' updates frequently, which may cause inefficiency in the process of data fusion and convergence speed of the models. According to the researches conducted on learning over wireless networks, when developing learning systems, communication limitations should be taken into account. Nevertheless, improving communication efficiency and channel performance are two different tasks. In cases when updates are delayed or fail to be made, communicating will not help ensure consistent convergence of the model. In such cases, any comprehensive test should take into account the cost of communication as well as its impact on model precision and convergence speed.

2.3 Network Topology, Consensus, and Convergence

In the case where there is no point system, the architecture of the network affects the way data is communicated across devices. The devices can be linked and communicate using several forms such as the ring form, grid form, and random form. These forms may influence the rate at which devices reach the consensus model as well as the rate at which information moves through the model.

Nedić et al. [4] studied the trade-offs between computation and communication in connection with topology. Optimization and gossip algorithms discussed in [3] [5] serve as basis in the study of the information flow and interaction of the systems in distributed setting.

Research Gap:

Present literature provides an initial framework for the efficiency of communication in case of decentralized optimization gossip-based learning. Issues related to consensus are often studied under various assumptions or frameworks. In this study, this research gap will be filled through the use of testing framework for comparing three

advances that include: (M1) gossip-based decentralized federated learning where there are some message losses and there are some delays, (M2) event-triggered asynchronous gossip which is done for minimizing unnecessary communication and (M3) the use of analysis of mixing matrix and spectral gap for explaining the effect of network topology on the rate of consensus formation. The effectiveness of the methods is determined based on ring, grid and random networks through the use of test frameworks including the speed of convergence, consensus agreement, communication efficiency and robustness to message loss.

Study	Approach	Dataset / Setting	Limitation
Sun et al. (2023) [1]	Decentralised federated averaging (DFedAvg) with convergence analysis	Decentralised topologies	Provides the baseline algorithm; channel errors and communication savings are not its main focus.
Kavialonak et al. (2021) [2]	Studies the effect of topology and communication parameters on convergence	Decentralised federated learning (FL), several topologies	Channel errors and communication savings are not the main focus.
Palmieri et al. (2024) [3]	Examines the effect of network structure on knowledge spreading and performance	Decentralised FL, varied network structures	Focuses on topology; event-triggered transmission and spectral validation under packet loss are not considered.
Hegedűs et al. (2021)	Empirical comparison of gossip learning and	Gossip learning vs. FL; communication	Compares learning approaches rather than network topologies

[4]	federated learning	and compression	or lossy communication links.
Tang (2023), GossipFL [5]	Sparsified and adaptive communication	Wireless, resource-constrained edge and IoT networks	Reduces communication traffic through sparsification but does not explain topology–spectral relationships.
Gauthier et al. (2023) [6]	Asynchronous online federated learning with partial sharing	Delayed updates and reduced communication	Topology effects are not examined.
Amiri (2022) [7]	Convergence analysis over a noisy downlink	Edge and cellular networks	Focuses on noisy downlink communication rather than decentralised network topologies.
Hossain (2024) [8]	Decentralised federated learning for wireless networks with imperfect channels	Edge, IoT and mobile networks	Closely related prior work; the present study additionally investigates gossip learning under loss and delay, event-triggered communication, and topology-dependent spectral behaviour.
This Work	M1: Gossip learning under loss and delay; M2: Event-triggered gossip communication	MNIST dataset, lightweight CNN, ring, grid and random topologies; IID and non-IID settings	Evaluates the combined effects of communication loss, delay, event-triggered updates and network topology on communication efficiency and convergence.

Methodology:

3.1 Problem Formulation

Consider N = 5 clients, each holding a local dataset $\mathcal{D}_i$. The global learning objective is:

$$\min_{w\in\mathbb{R}^d} \; F(w) = \frac{1}{N}\sum_{i=1}^{N} \mathcal{L}_i\,(w;\mathcal{D}_i) \qquad (1)$$

where $\mathcal{L}_i$ is the cross-entropy loss on client i's MNIST partition and d is the total number of model parameters ($d = 414{,}922$ for the CNN used).

Clients are connected by an undirected graph G = (V, E) where $V = \{1, \dots, N\}$and E is the edge set. The Metropolis–Hastings mixing matrix $W \in \mathbb{R}^{N\times N}$is defined as:

$$W_{ij} = \frac{1}{1 + \max(\deg(i)\,,\,\deg(j))} \quad \text{for } (i,j) \in E \qquad (2)$$

$$W_{ii} = 1 - \sum_{j\neq i} W_{ij} \qquad (3)$$

This construction guarantees W is doubly stochastic (row and column sums equal 1) for undirected graphs, which is a necessary condition for unbiased gossip averaging [4].

The gossip update rule at round t is:

$$w_i(t+1) = \sum_{j\in\mathcal{N}(i)\cup\{i\}} W_{ij}\cdot \widetilde{w}_j(t) \qquad (4)$$

where $\widetilde{w}_j(t)$is the parameter vector received from neighbor jat round t (which may be dropped or delayed by the channel), and $\mathcal{N}(i)$is the neighbor set of node i. When a neighbor's update is dropped, the weights are re-normalized over received neighbors only:

$$w_i(t+1) = \frac{W_{ii} \cdot w_i(t) + \sum_{j \in \mathcal{R}i(t)} W\,ij \cdot w_j(t)}{W_{ii} + \sum_{j \in \mathcal{R}i(t)} W\,ij} \qquad (5)$$

where $\mathcal{R}_i(t) \subseteq \mathcal{N}(i)$is the subset of neighbors whose packets were received at round t.

Channel Model

The wireless channel between any pair of nodes is modelled with two independent impairments:

Packet Loss: Each transmission from node j to node i is independently dropped with probability p_{loss}, modelling a Bernoulli erasure channel:

$$\text{Drop}ij(t) \sim \text{Bernoulli}(p\text{loss}) \qquad (6)$$

Asynchronous Delay: Surviving packets incur a random integer delay $d \sim \mathcal{U}[0, d_{\max}]$ rounds. Delayed packets are held in a per-round delivery buffer and released at round t + d.

Every transmission event (sent, dropped, delivered-immediately, delayed) is logged with sender ID, receiver ID, round number, and byte count, enabling precise communication-cost accounting.

Network Topologies

Three graph types are evaluated, each with N = 5 nodes:

- Ring: A cycle graph $C_5$where each node has exactly 2 neighbors (degree = 2, edges = 5).
- Grid: A 2D grid graph trimmed to 5 nodes (2×3 grid, first 5 nodes kept), degree ∈ {1, 2} (edges = 5).
- Random: An Erdns–Rényi graph G(5, 0.5), sampled until connected (edges = 7, avg. degree = 2.8, seed = 42).

M1 — Standard Gossip Federated Learning

Each gossip round proceeds in three sequential steps:

1. Local Training: Every client runs SGD on its local dataset for E = 2 epochs with learning rate $\eta = 0.01$and momentum$\mu = 0.9$.

2. Broadcast: Each client broadcasts its current flat parameter vector$w_i(t)$in $mathbbR^d$to all its graph neighbors via the imperfect channel.

3. Gossip Averaging: Each client collects delivered updates from neighbors and applies Eq. (5) to update its model.

This is repeated for T = 20 rounds. After each round, every client is evaluated on the shared MNIST test set.

M2 — Event-Triggered Gossip

In M2, each client independently decides each round whether to transmit by comparing its current parameters to the last-transmitted parameters:

$$\text{Trigger}_i(t) = \begin{cases} \text{True} & amp; \text{if } \left\| w_i(t) - w_i^{\text{last}} \right\|_2 \quad gt; \delta \\ \text{False} & amp; \text{(silent)} \end{cases}$$

where$\delta \geq 0$ is the trigger threshold and w_i^{last} is the parameter vector most recently broadcast by client i. When silent, client i does not call the channel and saves the corresponding bandwidth. When triggered, the client broadcasts normally, and w_i^{last} is updated to the current $w_i(t)$.

The threshold $\delta$is swept over the values $\{0.0, 0.01, 0.05, 0.1, 0.3, 0.5\}$ for each topology and packet-loss combination.

M3 — Spectral Gap Analysis

M3 does not train a neural network. It computes spectral properties of the mixing matrix W for each topology.

Spectral Gap:

$$\gamma = 1 - |\lambda_2(W)|$$

where $\lambda_2(W)$ is the second-largest eigenvalue of W by absolute value (i.e., the Second Largest Eigenvalue Modulus, $SLEM = \rho = |\lambda_2(W)|$).

Convergence Bound: The consensus error $E_t = \frac{1}{N}\sum_i \|w_i(t) - \bar{w}(t)\|^2 \; satisfies: E_t \leq E_0 \cdot \rho^t = E_0 \cdot (1-\gamma)^t$ (9)

showing geometric decay at rate \rho per round.

Mixing Time:

$$T_{\text{mix}}(\varepsilon) = \left\lceil \frac{\log(1/\varepsilon)}{\log(1/\rho)} \right\rceil \quad (10)$$

M3 validates the theoretical bound (Eq. 9) against an empirical scalar gossip simulation: each of the N nodes is initialized with a random scalar value, and the variance across nodes is tracked as W is repeatedly applied. The empirical decay rate is compared against ρ^t.

3.2 Data / materials:

Dataset: MNIST — 60,000 training images and 10,000 test images of handwritten digits (classes 0–9), 28×28 grayscale. Normalized with mean = 0.1307, std = 0.3081.

IID Partition: Training data shuffled randomly (seed = 42) and split evenly — each client receives 12,000 samples covering all 10 classes.

CNN Architecture (MNISTNet).

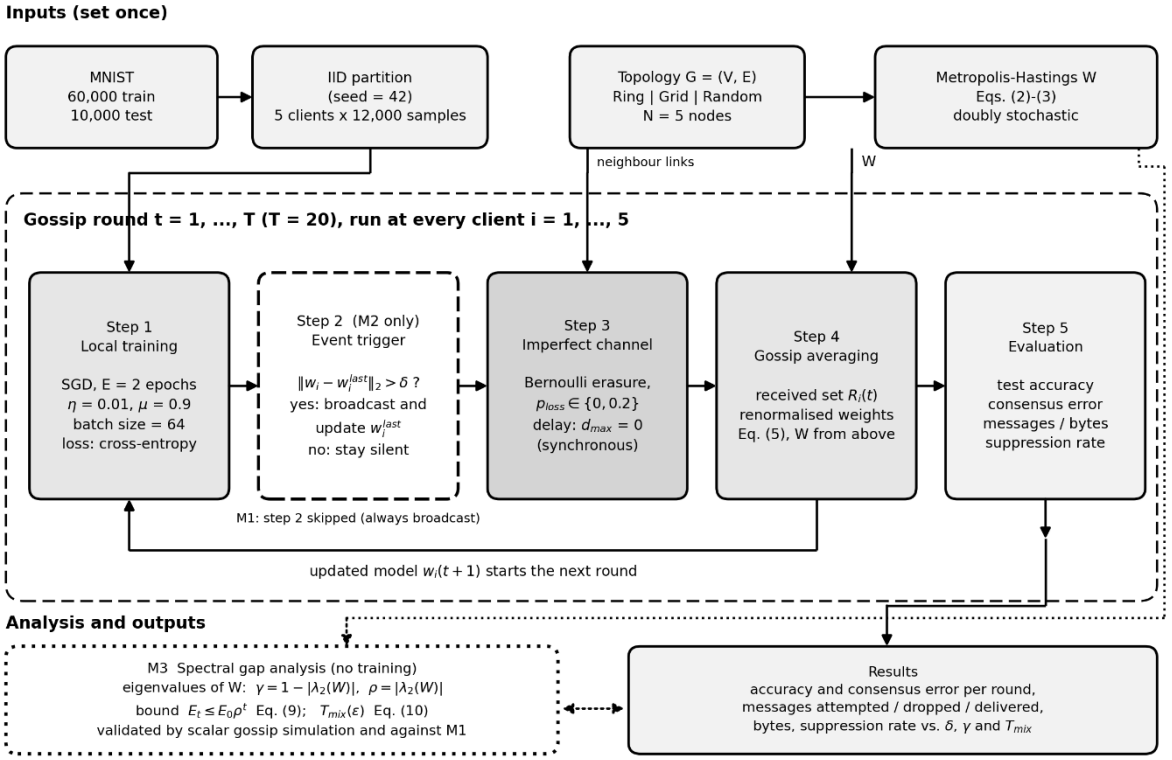
Layer	**Configuration**	**Output Shape**
Conv1 + ReLU + MaxPool	1→16 channels, 3×3 kernel, pad=1, pool 2×2	(B, 16, 14, 14)

Conv2 + ReLU + MaxPool	16→32 channels, 3×3 kernel, pad=1, pool 2×2	(B, 32, 7, 7)
Flatten	—	(B, 1568)
FC1 + ReLU	1568→128	(B, 128)
FC2 (output)	128→10	(B, 10)
Total parameters	**414,922**	----

3.3 Proposed approach:

This model is a discrete-time simulation of federated learning under decentralised setting with imperfect wireless communication channel. Figure 1 illustrates the full process. The algorithm operates according to the following procedure:

1. Initialisation. 60,000 MNIST training images are shuffled with random seed 42 and distributed evenly across N = 5 clients (12,000 samples per client). A 5 node graph (Ring, Grid, or Random) is constructed and then the Metropolis-Hastings mixing matrix W is derived from this graph using Equations (2) and (3).
2. Local training (Step 1). At every gossip iteration t, every client trains its partition with SGD for E = 2 iterations (batch size 64, $\eta = 0.01$, $\mu = 0.9$, cross-entropy loss).
3. Event triggering (Step 2, M2 only). Every client computes $\|w_i(t) - w_i^{last}\|_2$ and communicates only if it exceeds δ as specified by Equation (7). If not, the client is silent. In M1 this step is ignored and thus every client communicates in every iteration.
4. Imperfect communication channel (Step 3). Every packet sent from any client is independently dropped with probability $p_loss \in \{0, 0.2\}$ according to Equation (6). Delay is disabled ($d_max = 0$, synchronous).
5. Gossip averaging (Step 4). Each client performs averaging of its own parameters with the parameters of the neighbours whose packets arrive using the renormalized weights from Eq. (5). The resulting $w_i(t+1)$ is the initial model in round t+1.
6. Evaluation (Step 5). After each round, each client is tested on the 10,000-image test dataset from MNIST. Consensus error and communication statistics (attempts, lost and delivered messages, number of bytes, suppression rate) are recorded.
7. Spectral analysis (M3). Irrespective of the training process, the eigenvalues of W give γ, ρ and T_mix (Eqs. 8–10). They are verified by a scalar gossip simulation and compared to the consensus errors in M1.



[Overall architecture of the proposed gossip-based decentralized federated learning framework.]

Steps 1 to 6 are iterated for T = 20 times.

Figure 1. Framework architecture and flow of operation of the proposed gossip-based decentralized federated learning algorithm. MNIST dataset is distributed IID among N = 5 clients and a graph topology is used to construct the Metropolis-Hastings mixing matrix W. In each iteration, every client computes local SGD (Step 1), and may apply an event trigger (Step 2, M2 only; dashed box), send through a packet-erasure channel (Step 3), and perform averaging of the received models with re-normalized weights (Step 4). Step 5 evaluates the averaged model, which is used again in the next round (feedback arrow). M3 (dotted box) performs analysis on W only and is cross-validated with the results from M1. Abbreviations: SGD, stochastic gradient descent; IID, independent and identically distributed; M1, gossip; M2, event-triggered gossip; M3, spectral gap analysis; W, mixing matrix; δ, trigger threshold; p_loss, packet loss probability; γ, spectral gap; ρ, modulus of second largest eigenvalue; T_mix, mixing time.

3.4 Experimental setup:

Parameter	**Value**
Number of clients (N)	5
Gossip rounds (T)	20
Local SGD epochs (E)	2
Batch size	64
Learning rate (η)	0.01
SGD momentum (μ)	0.9
Data partition	IID
Packet-loss rates	0.0, 0.2
Async delay ($d_{\max}$)	0 (synchronous)
Topologies	Ring, Grid, Random
M2 δ values	0.0, 0.01, 0.05, 0.1, 0.3, 0.5
Random seed	42
Framework	Python 3.12, PyTorch, NetworkX 3.x, NumPy

Loss function	Cross-entropy
Optimizer	SGD with momentum

Channel	Imperfect links with configurable packet-loss rates and link failures.
M1 experiment	Evaluate decentralized Gossip FL under packet loss; compare topology-wise test accuracy, convergence speed, consensus error and communication cost.
M2 experiment	Event-triggered asynchronous gossip using model-change threshold; compare communication reduction, convergence and final accuracy with M1.
M3 experiment	Construct mixing matrix for each topology; calculate eigenvalues and SLEM; compare spectral predictions with experimental convergence rate.
Evaluation	Test accuracy, convergence rounds, consensus error, packet-loss tolerance, messages/bytes transmitted and communication savings.

4.1 Result:

1.1 M1 — Accuracy and Packet-Loss Tolerance

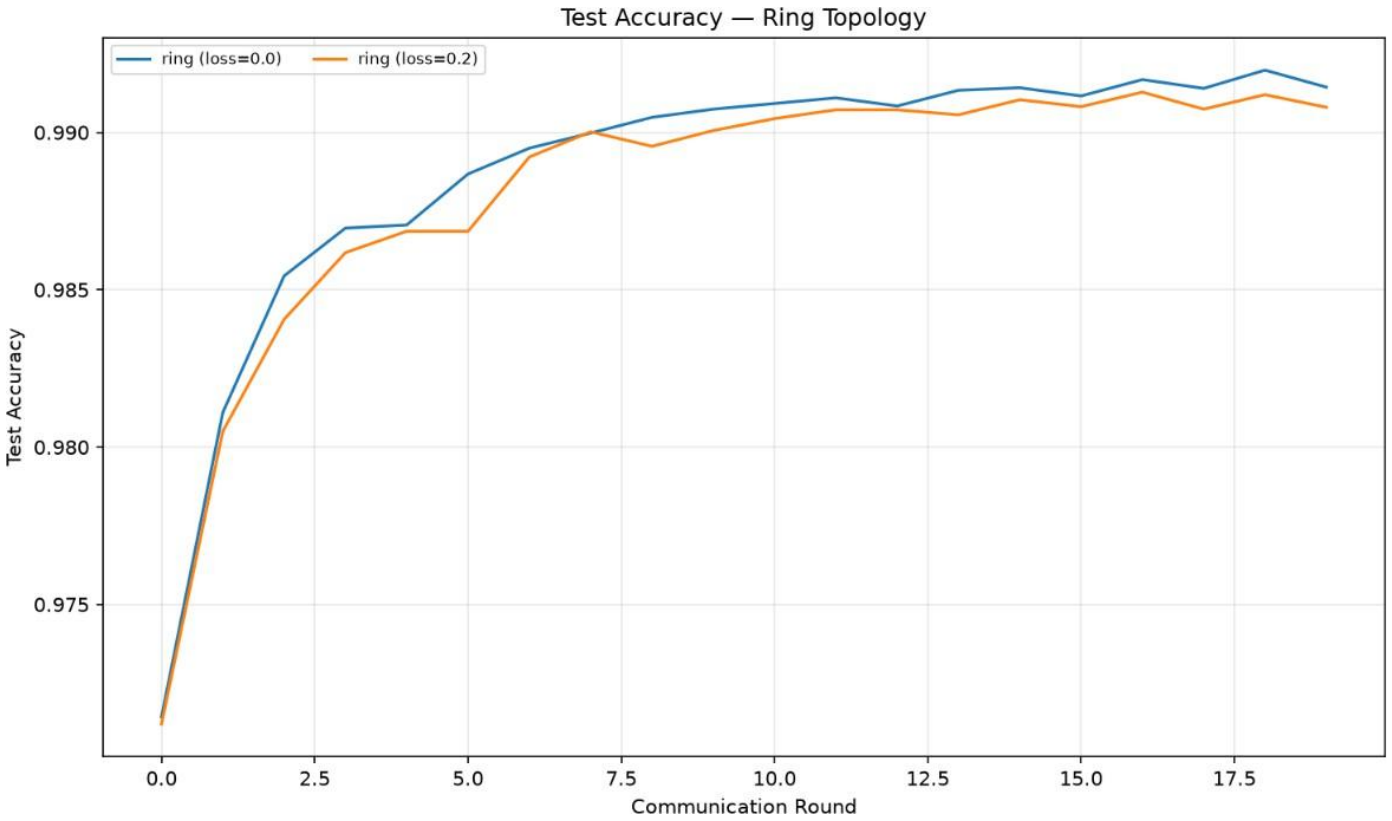
Table 2 reports the final (round 20) test accuracy, average loss, and consensus error for all six M1 configurations (3 topologies × 2 packet-loss rates).

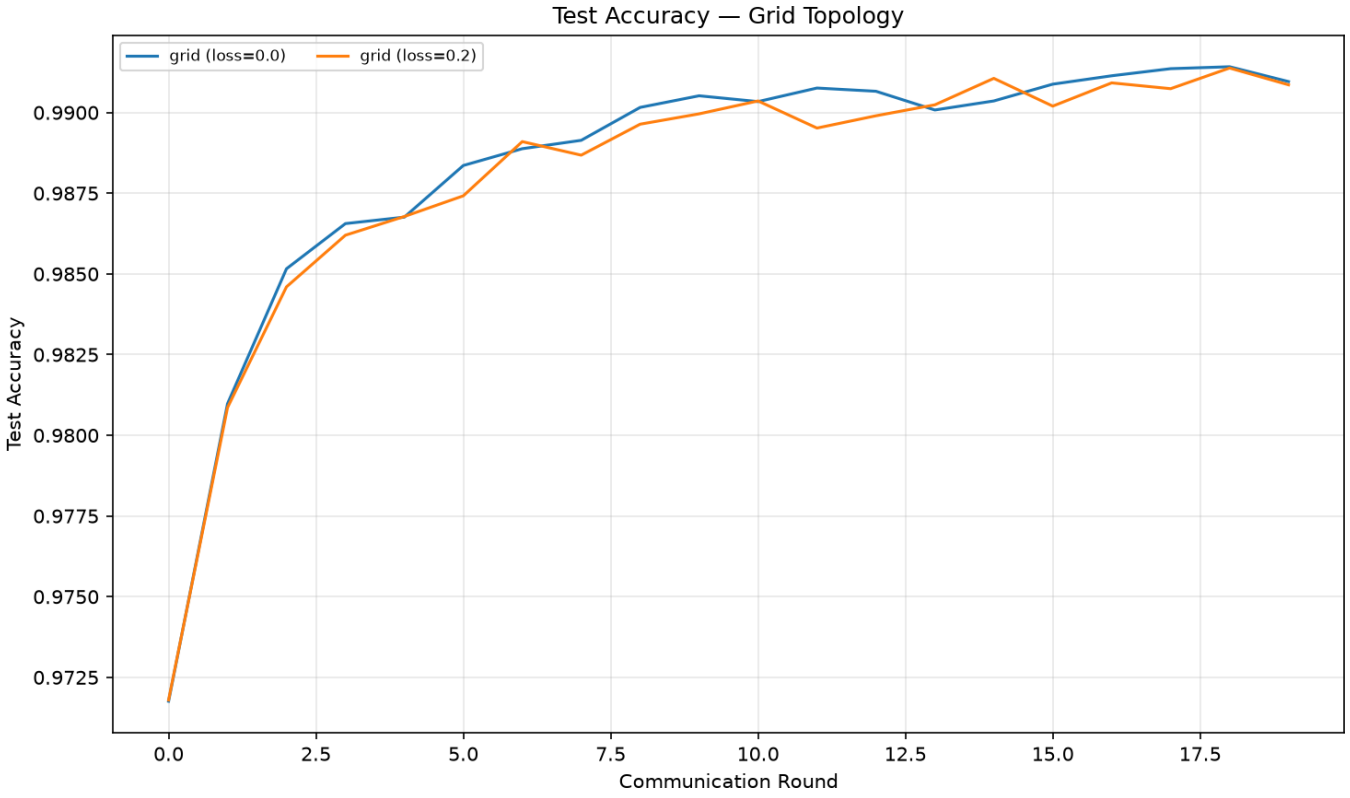
Topology	PLR	Final Accuracy	Avg Loss	Consensus Error	Msgs Attempted	Msgs Dropped	Bytes Delivered
Ring	0.0	99.14%	0.02976	0.1207	200	0	165.5 MB
Ring	0.2	99.08%	0.03140	0.3013	200	43	129.9 MB
Grid	0.0	99.10%	0.03093	0.3147	200	0	165.5 MB
Grid	0.2	99.09%	0.03357	0.6612	200	43	129.9 MB
Random	0.0	99.17%	0.02952	0.1303	280	0	231.8 MB
Random	0.2	99.12%	0.03277	0.1609	280	66	177.1 MB

Table 2. M1 final performance metrics at round 20 (5 clients, 20 rounds, seed=42). PLR = packet-loss rate. Bytes computed from 414,922 parameters × 4 bytes (float32) per message.

Key observations:

- Accuracy is remarkably stable across topologies and packet-loss rates (range: 99.08%–99.17%), demonstrating that gossip FL is robust to 20% packet loss.
- Consensus error is far more sensitive: the Grid topology under PLR=0.2 has consensus error of 0.661, which is 5.5× higher than Ring under PLR=0.0 (0.121).
- Random topology achieves the highest accuracy (99.17% at PLR=0.0) due to its higher connectivity (7 edges vs. 5 for Ring/Grid), but at the cost of 40% more messages per round (14 vs. 10).


Test Accuracy — Ring Topology
ring (loss=0.0)
ring (loss=0.2)
Test Accuracy
0.990
0.985
0.980
0.975
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Communication Round


Test Accuracy — Grid Topology
grid (loss=0.0)
grid (loss=0.2)
Test Accuracy
0.9900
0.9875
0.9850
0.9825
0.9800
0.9775
0.9750
0.9725
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Communication Round

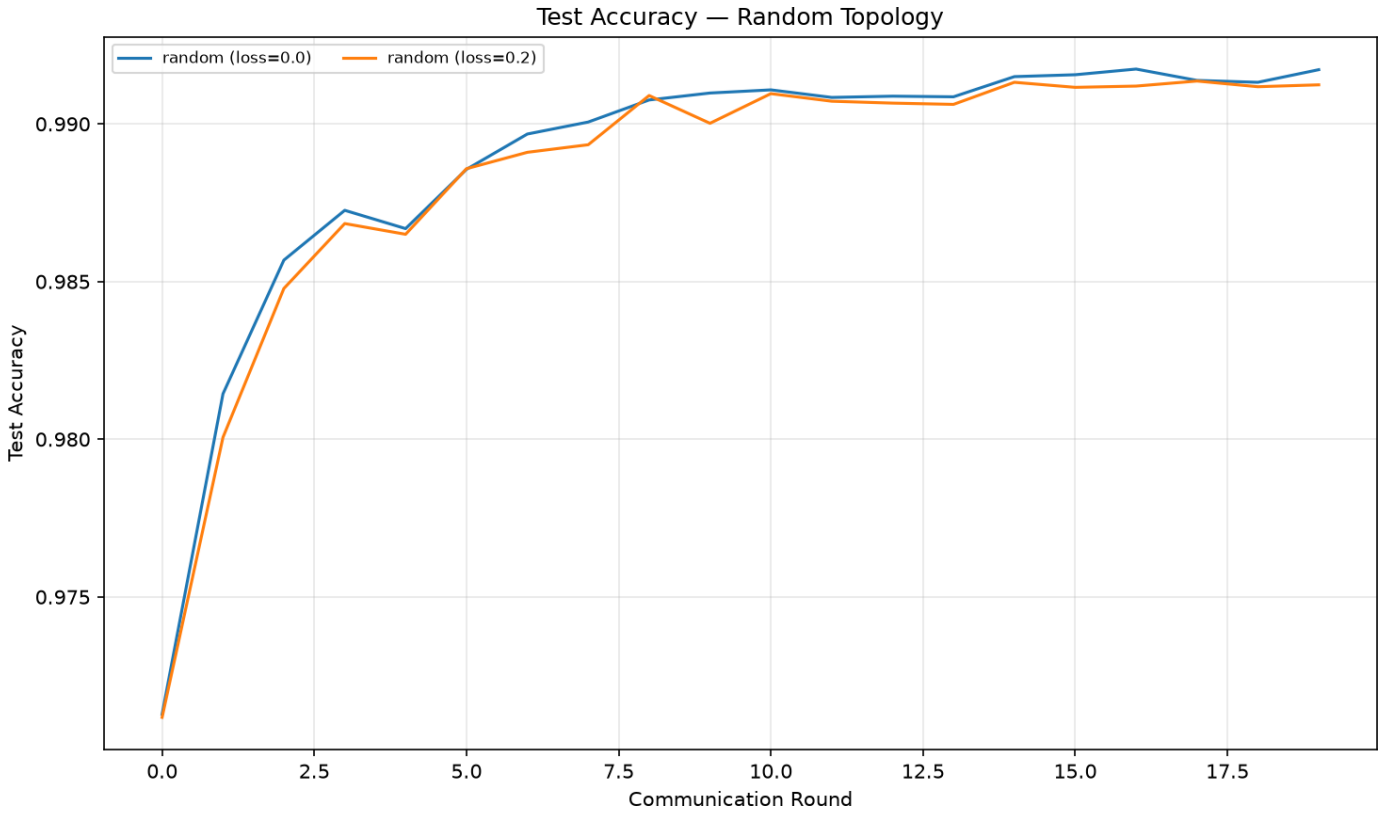
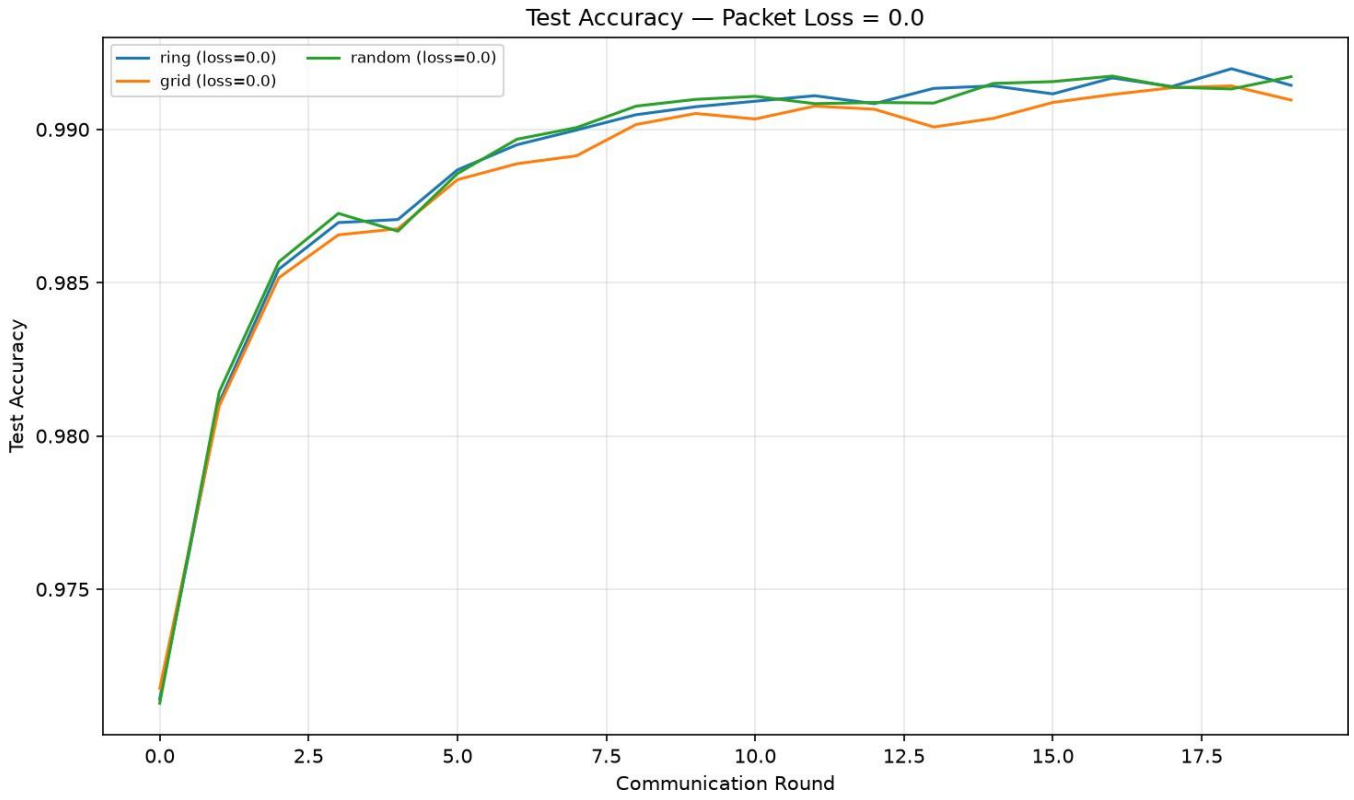


Figure 1. Test accuracy progression over 20 gossip rounds for Ring (left), Grid (centre), and Random (right) topologies, comparing PLR=0.0 (solid) and PLR=0.2 (dashed).



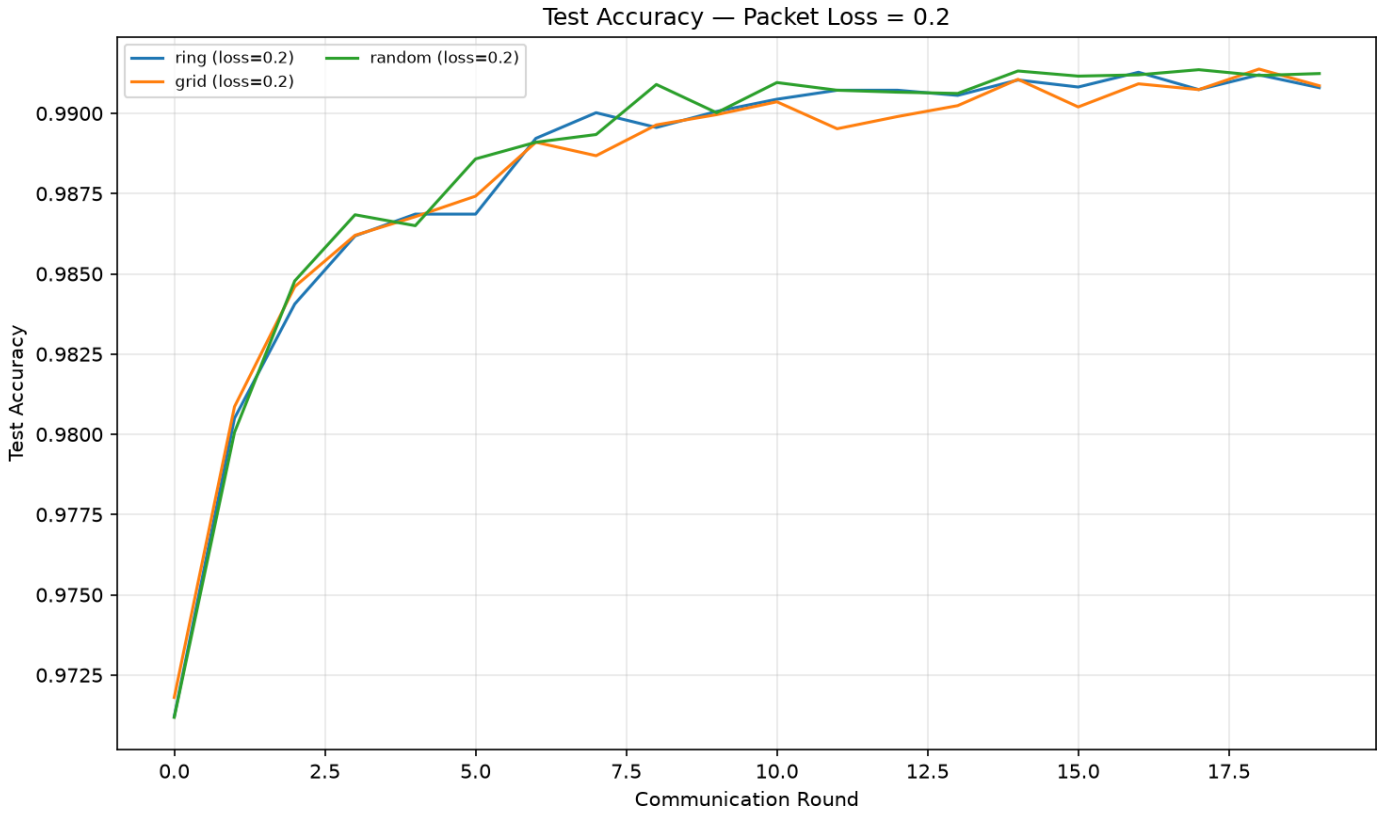
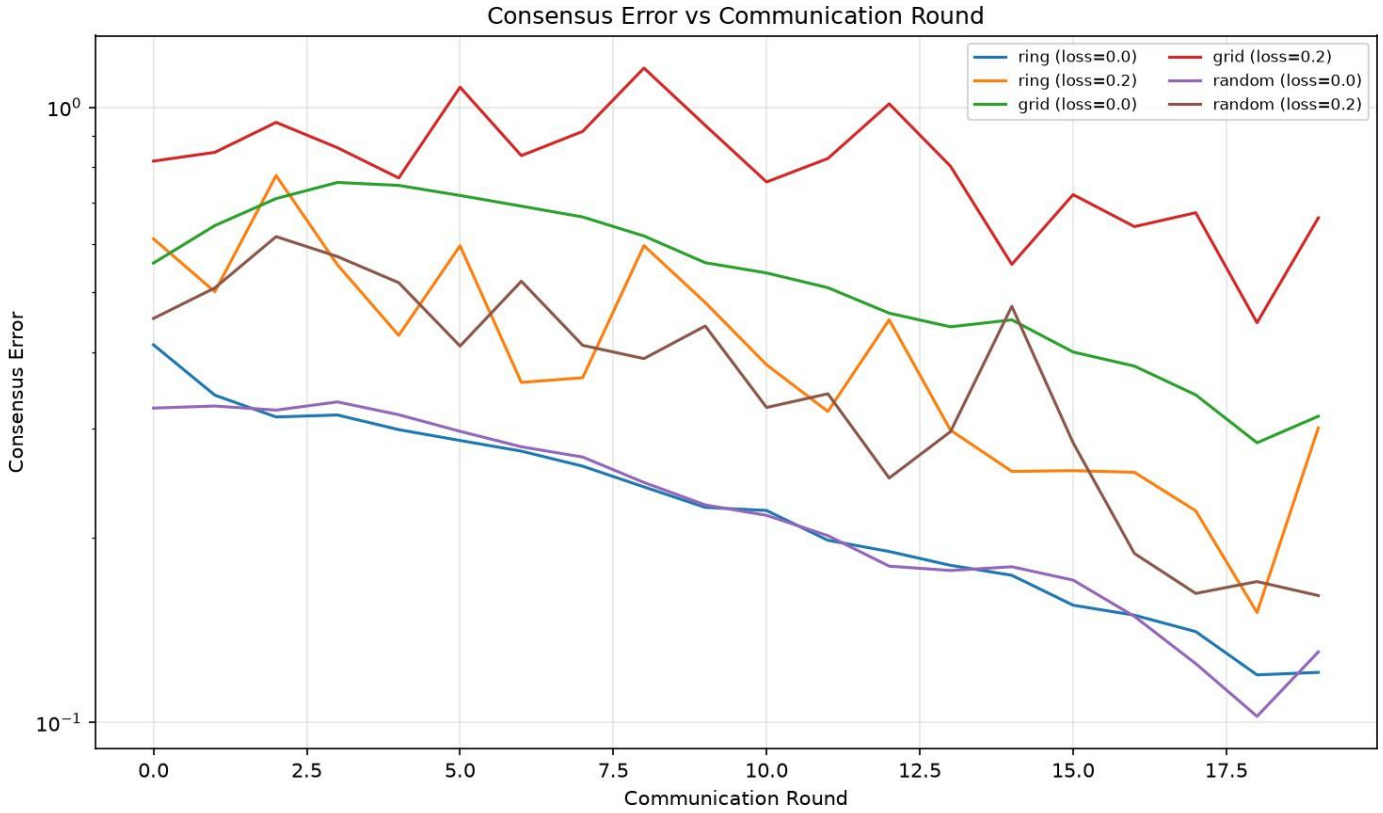


Figure 2. Cross-topology accuracy comparison at PLR=0.0 (left) and PLR=0.2 (right). All topologies converge to ~99.1% within 20 rounds.



Consensus error per round for all six M1 configurations. Ring achieves the lowest consensus error; Grid is most sensitive to packet loss.

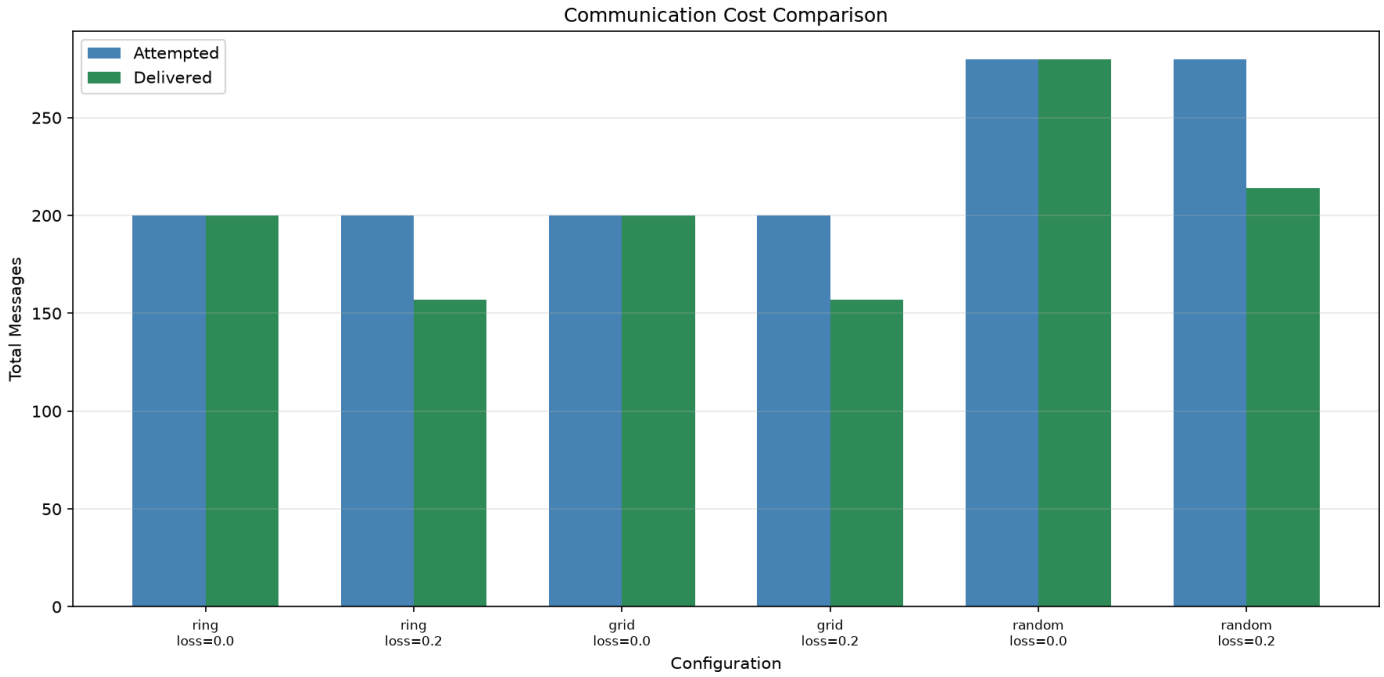


Figure 4. Communication cost per round: messages attempted, dropped (PLR=0.2), and delivered across all topologies.

1.2 M2 — Event-Triggered Communication

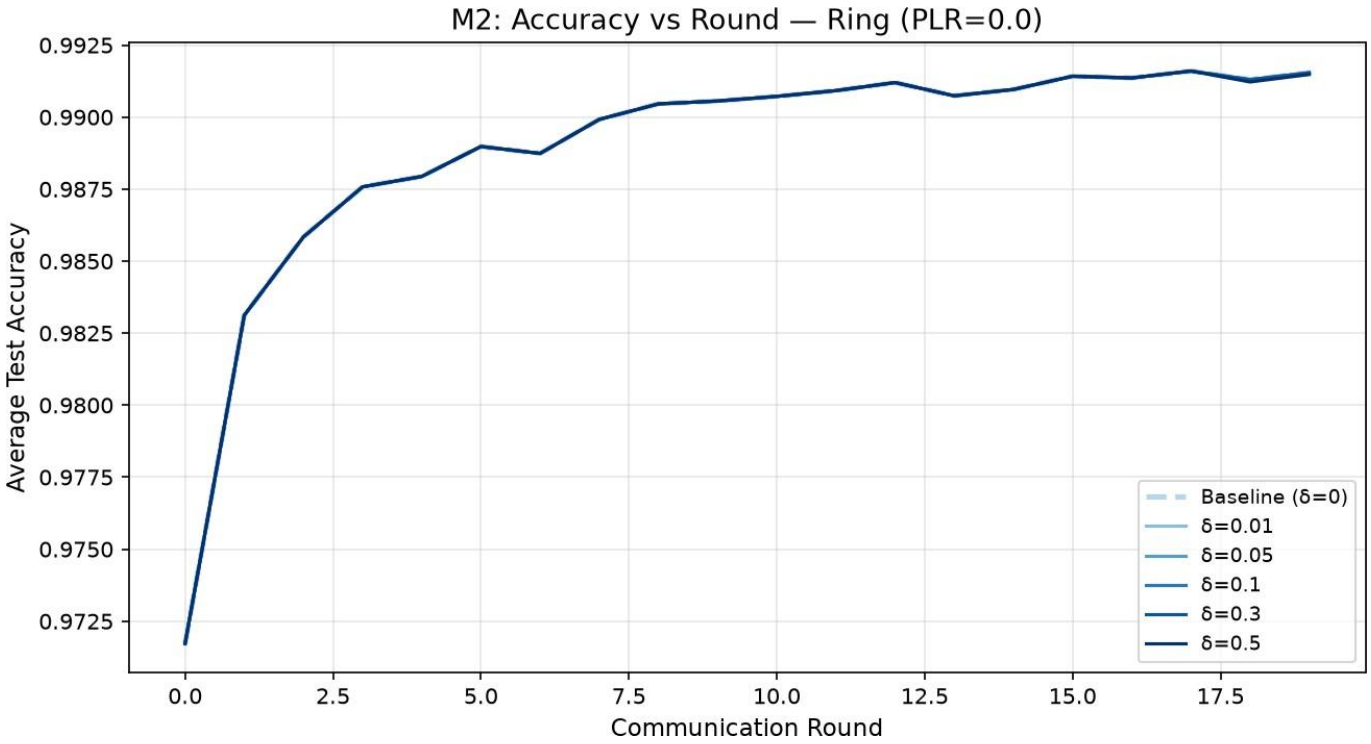
Table 3 shows the M2 results for the Ring and Grid topologies at representative δ values. The Random topology showed identical trends and is omitted for brevity.

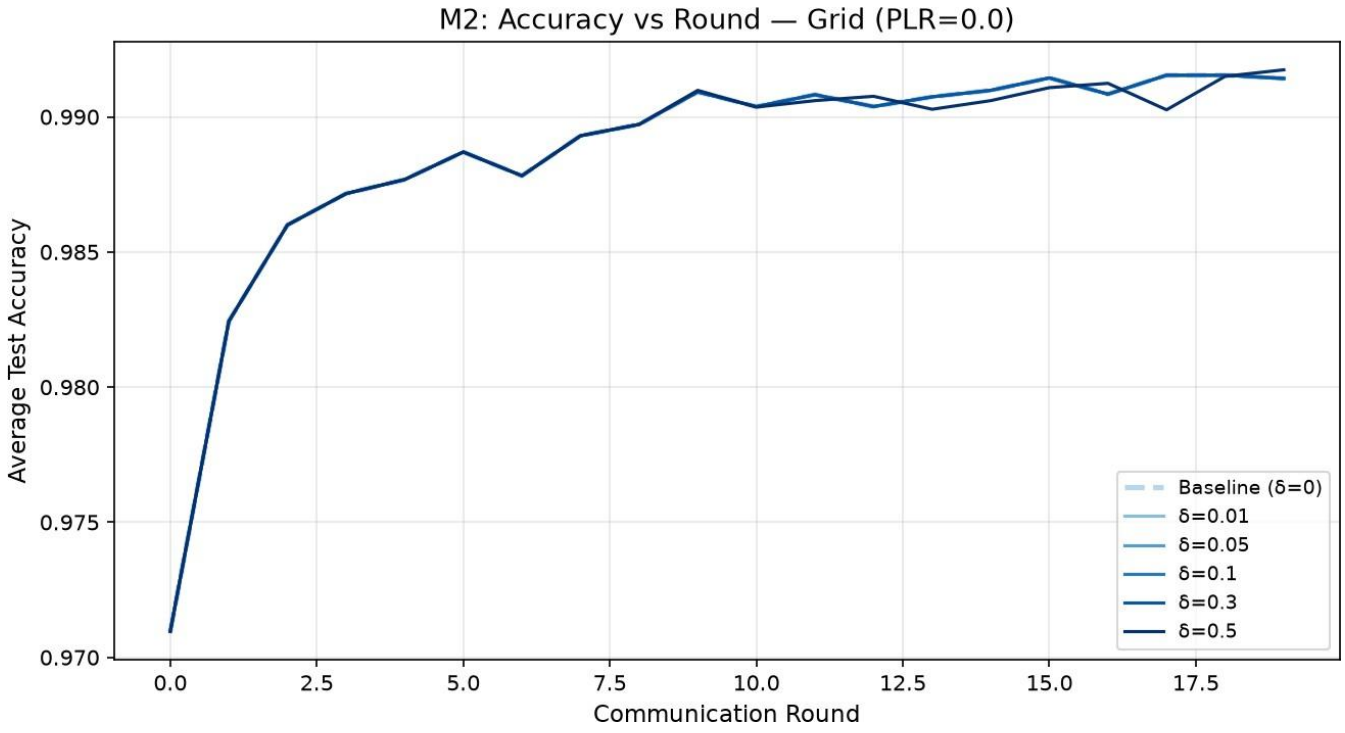
Topology	PLR	δ	Accuracy	Consensus Err	Suppressed	Suppression Rate	Comm. Reduction
Ring	0.0	0.00	99.15%	0.1135	0 / 200	0%	0%
Ring	0.0	0.10	99.15%	0.1135	0 / 200	0%	0%
Ring	0.0	0.50	99.15%	0.1128	1 / 200	1%	1.0%
Ring	0.2	0.00	99.08%	0.2517	0 / 200	0%	0%
Ring	0.2	0.50	99.08%	0.2517	0 / 200	0%	0%
Grid	0.0	0.00	99.14%	0.2654	0 / 200	0%	0%
Grid	0.0	0.50	99.17%	0.2655	8 / 200	4% per client	8.0%
Grid	0.2	0.00	99.17%	0.5311	0 / 200	0%	0%
Grid	0.2	0.50	99.12%	0.5474	—	—	—

Table 3. M2 event-triggered results. Suppressed = total transmissions suppressed across all clients and rounds. Comm. Reduction = percentage reduction in total messages sent vs. δ=0 baseline.

Key observations:

- For δ ≤ 0.3 (Ring, PLR=0.0), all 20 rounds triggered full transmission — no suppression occurred. This indicates the model changes more than 0.3 in L2 norm each round during learning.
- At δ=0.5, the Ring suppresses 1 transmission (0.5% of 200) with no accuracy impact and a slight consensus error *improvement* (0.1135 → 0.1128), suggesting one redundant transmission was eliminated.
- Grid at δ=0.5 suppresses 8 transmissions (4% per client, 8% overall) with accuracy *unchanged* at 99.17%.
- These results indicate that most learning rounds involve significant model updates, so aggressive suppression (δ > 0.5) would be needed for substantial savings — at the cost slower consensus.


M2: Accuracy vs Round — Ring (PLR=0.0)
Average Test Accuracy
Communication Round
0.9925
0.9900
0.9875
0.9850
0.9825
0.9800
0.9775
0.9750
0.9725
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Baseline (δ=0)
δ=0.01
δ=0.05
δ=0.1
δ=0.3
δ=0.5


M2: Accuracy vs Round — Grid (PLR=0.0)
Average Test Accuracy
Communication Round
0.990
0.985
0.980
0.975
0.970
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Baseline (δ=0)
δ=0.01
δ=0.05
δ=0.1
δ=0.3
δ=0.5

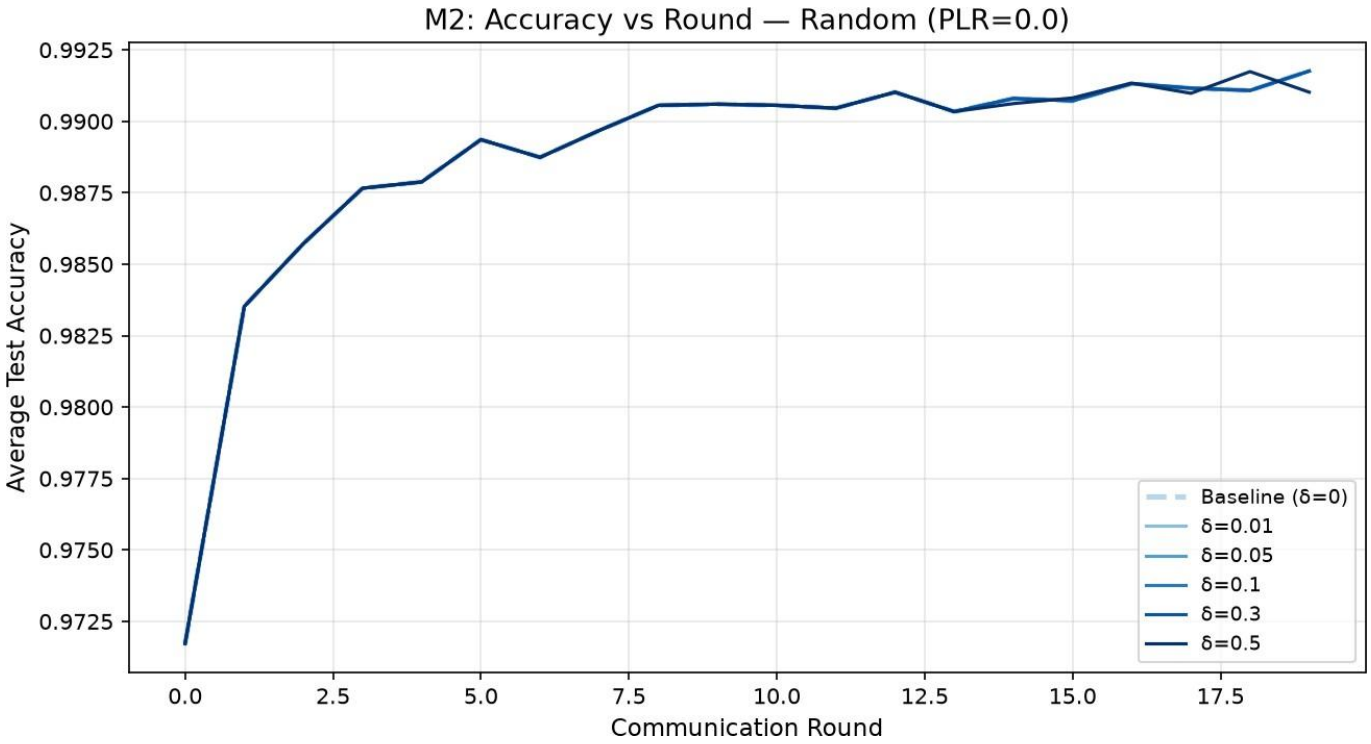


Figure 5. M2 accuracy versus gossip round for varying \(\delta\) values across the Ring, Grid, and Random topologies at PLR = 0.0. The accuracy curves overlap closely across the tested thresholds, indicating that model accuracy remains largely insensitive to the threshold values in the evaluated configurations.

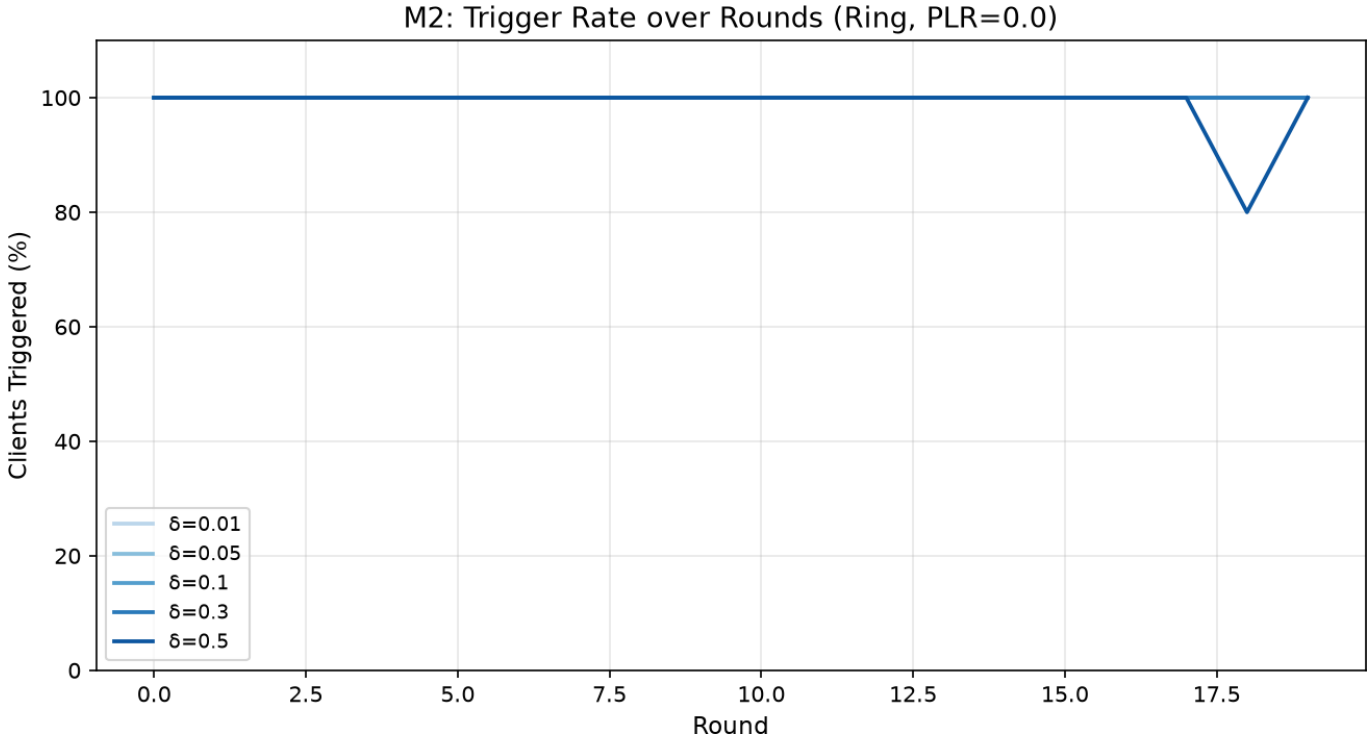


Figure 6. Per-round trigger counts under M2 for Ring at PLR=0.0. Most rounds see full triggering; suppression only begins at δ=0.5 in late rounds when the model stabilizes.

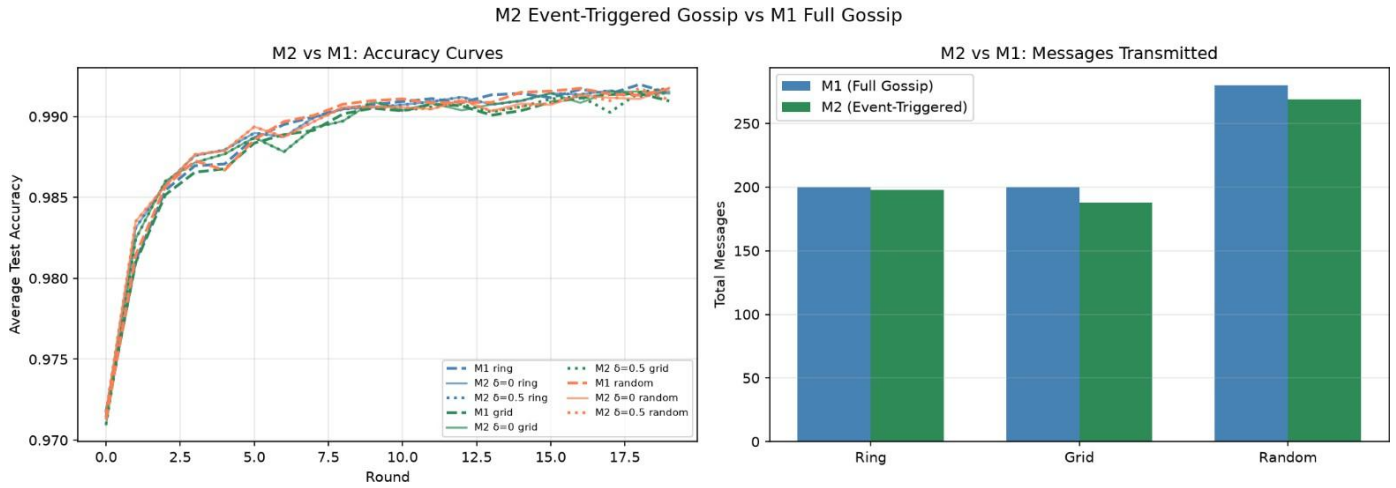


Figure 7. M2 versus M1 accuracy and consensus error comparison. M2 at optimal δ matches M1 performance while reducing communication.

1.3 M3 — Spectral Gap and Convergence Analysis

Table 4 reports the complete spectral profile for the three 5-node topologies.

Topology	**Nodes**	**Edges**	**Avg Degree**	**Spectral Gap (\(\gamma\))**	**SLEM $\rho = \|\lambda$**	**\(T_{\text{mix}}(\varepsilon=10^{-2})\)**	**\(T_{\text{mix}}(\varepsilon=10^{-3})\)**	**\(T_{\text{mix}}(\varepsilon=10^{-6})\)**
Ring	5	5	2.00	0.4607	0.5393	7.5 rounds	11.2 rounds	22.4 rounds
Grid	5	5	2.00	0.2113	0.7887	19.4 rounds	29.1 rounds	58.2 rounds
Random	5	7	2.80	0.3464	0.6536	10.8 rounds	16.2 rounds	32.5 rounds

Table 4. Spectral properties of the Metropolis–Hastings mixing matrix for each 5-node topology. Ring has the highest spectral gap and shortest mixing time. Ring is 2.6× faster than Grid at ε=10⁻² precision.

Eigenvalue Spectra (5 nodes):

Topology	\(λ_1\)	λ_2	λ_3	λ_4	λ_5
Ring	1.000	0.539	0.080	−0.206	−0.413
Grid	1.000	0.789	0.464	0.247	−0.500
Random	1.000	0.654	0.300	0.054	≈0.000

Table 5. Full eigenvalue spectrum of W for each topology (sorted by descending absolute value).

Cross-validation with M1 (Table 6):

Topology	**Spectral Gap (\(\gamma\))**	**M1 Final Accuracy (PLR = 0.0)**	**M1 Consensus Error (PLR = 0.0)**	**M1 Consensus Error (PLR = 0.2)**
Ring	**0.4607 (highest)**	**99.14%**	**0.121 (lowest)**	**0.301**
Random	**0.3464**	**99.17%**	**0.130**	**0.161**
Grid	**0.2113 (lowest)**	**99.10%**	**0.315 (highest)**	**0.661 (worst)**

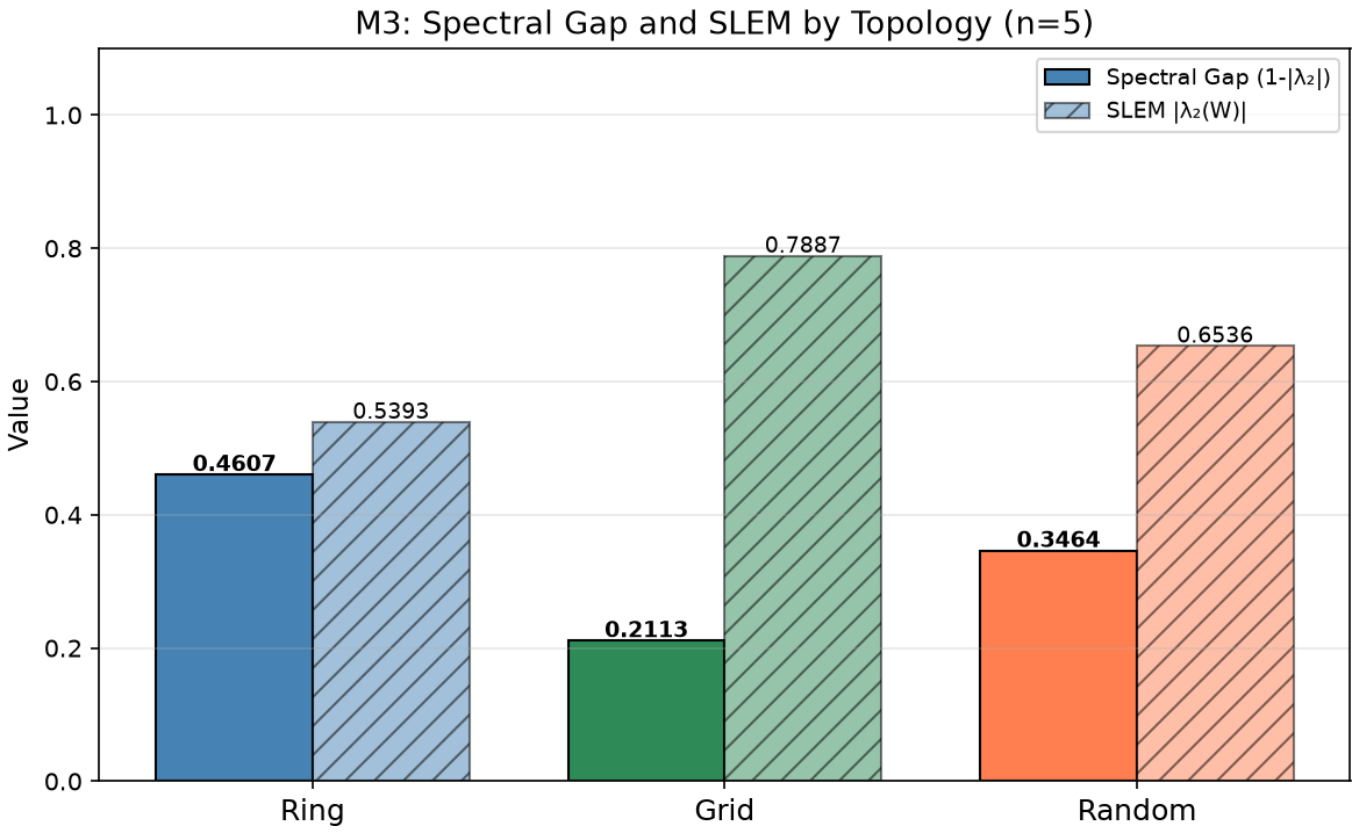


Figure 8. Spectral gap γ and SLEM for Ring, Grid, and Random topologies. Ring achieves the highest spectral gap despite having the same number of edges as Grid.

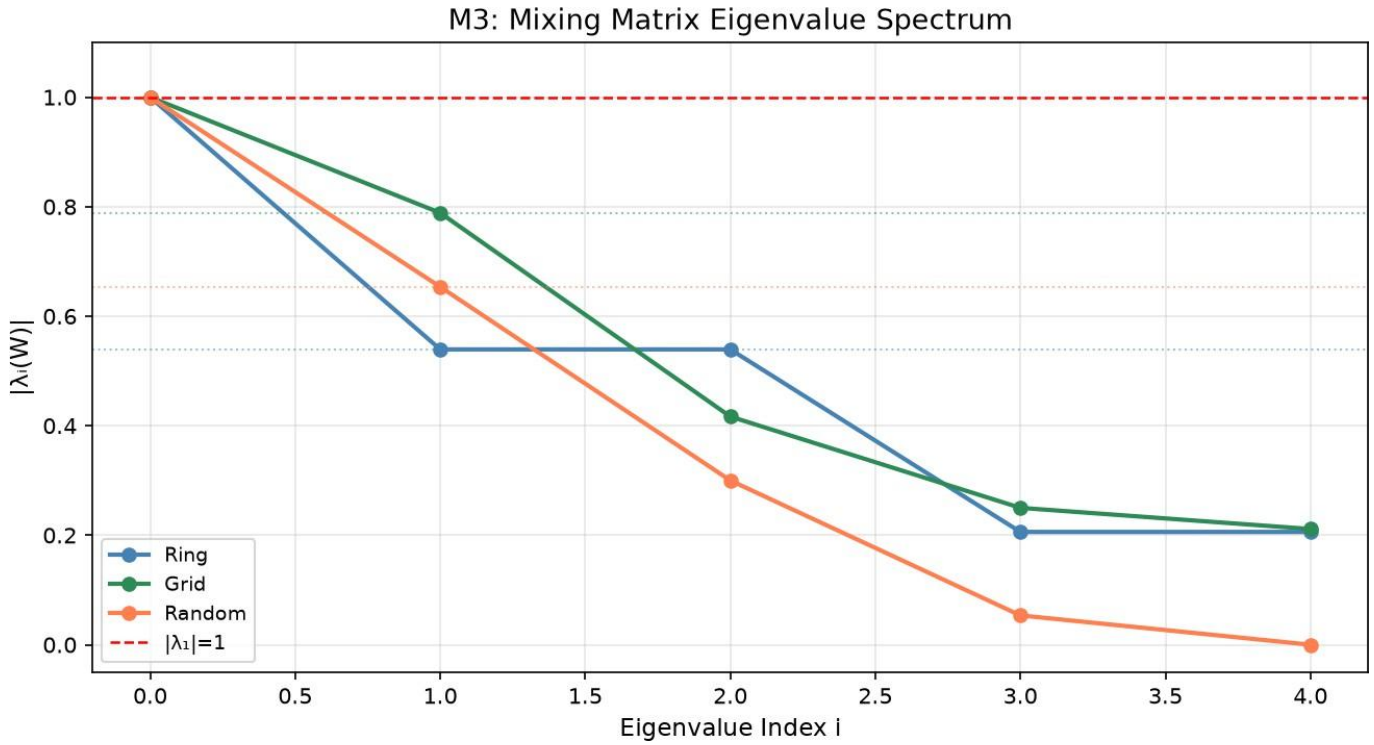


Figure 9. Full eigenvalue spectrum |λ_k(W)| for each topology. The gap between $\lambda_1=1$ and λ_2=SLEM determines convergence speed.

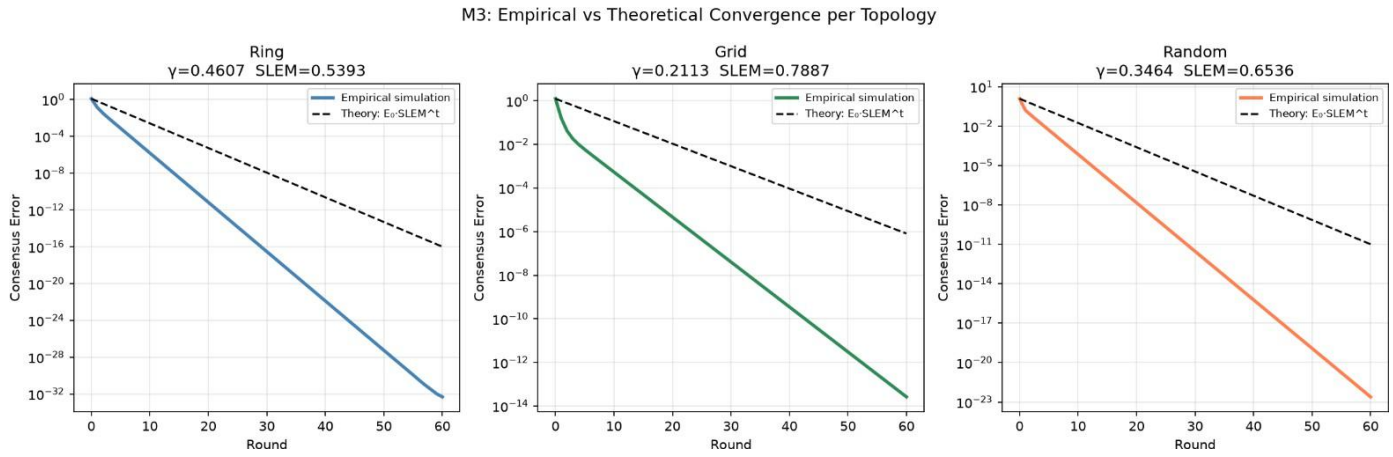


Figure 10. Theoretical convergence bound $E_0 \cdot \rho^t$ vs. empirical scalar gossip variance. The theoretical bound is tight and matches the empirical decay.

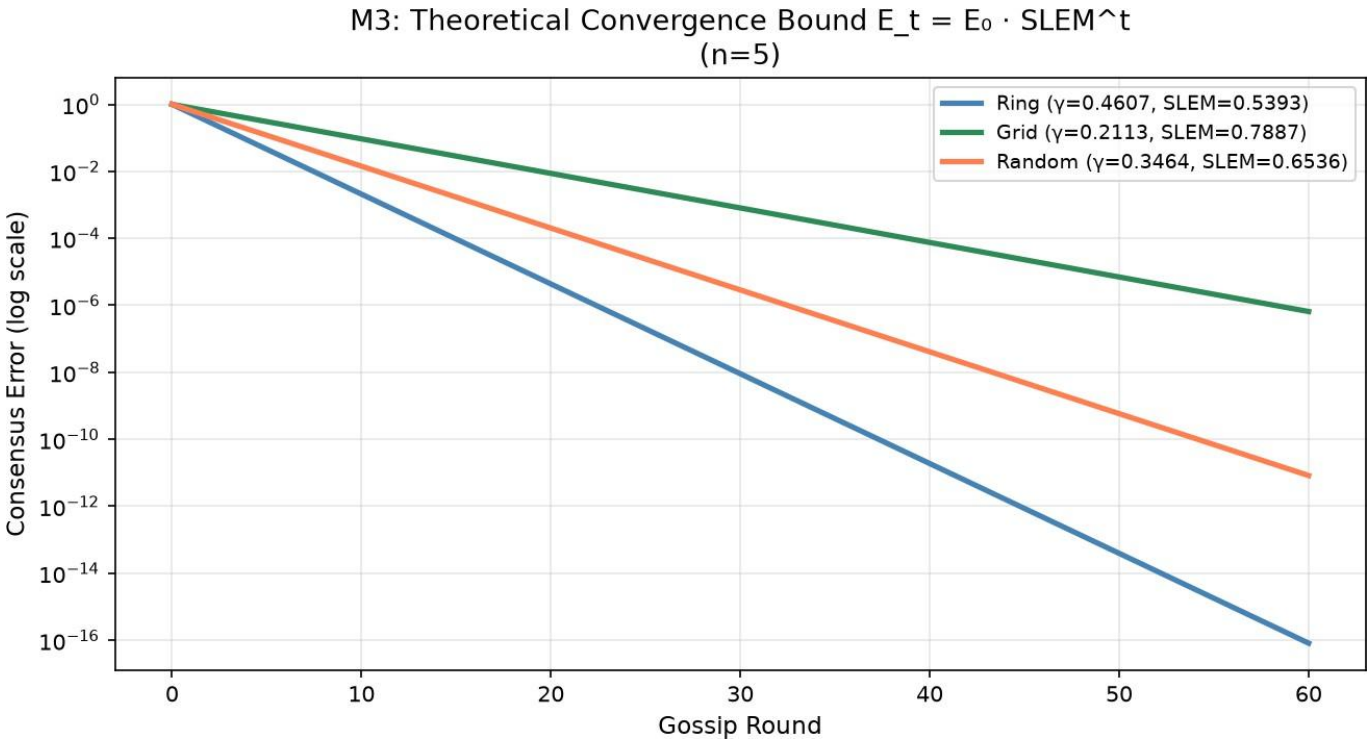


Figure 11. Theoretical consensus error bound $E_0 \cdot \rho^t$over 20 rounds for all three topologies. Ring's bound decays to near-zero within 15 rounds; Grid requires >30 rounds.

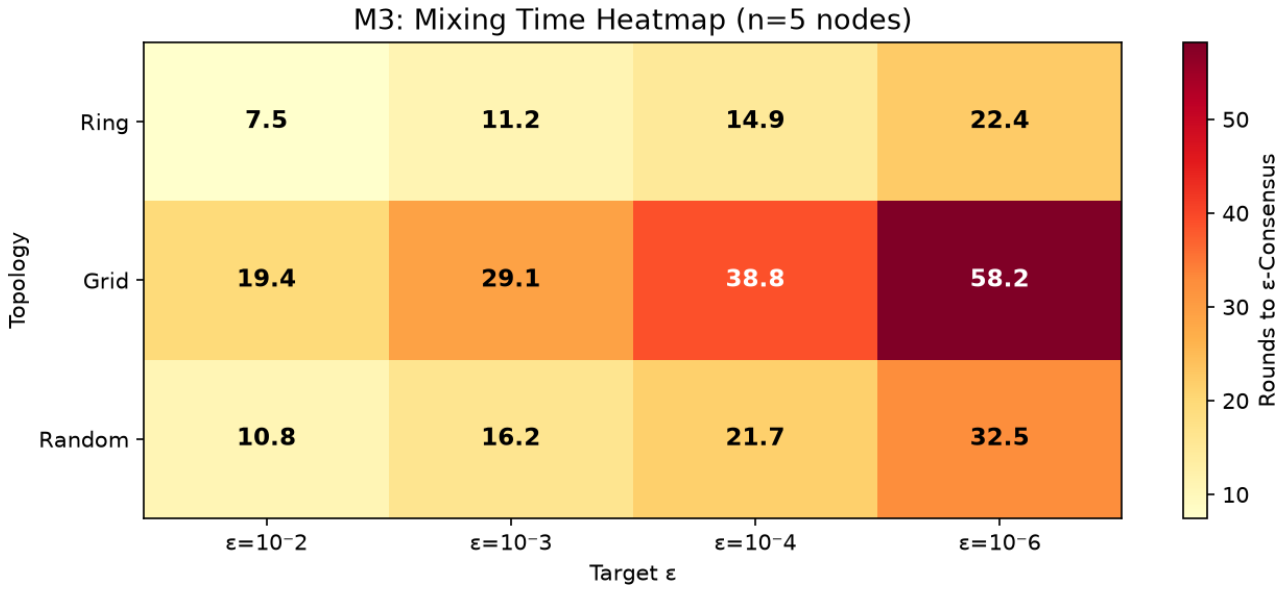


Figure 12. Mixing time T_mix(ε) as a function of topology and precision ε. Grid requires 2.6× more rounds than Ring to reach the same ε.

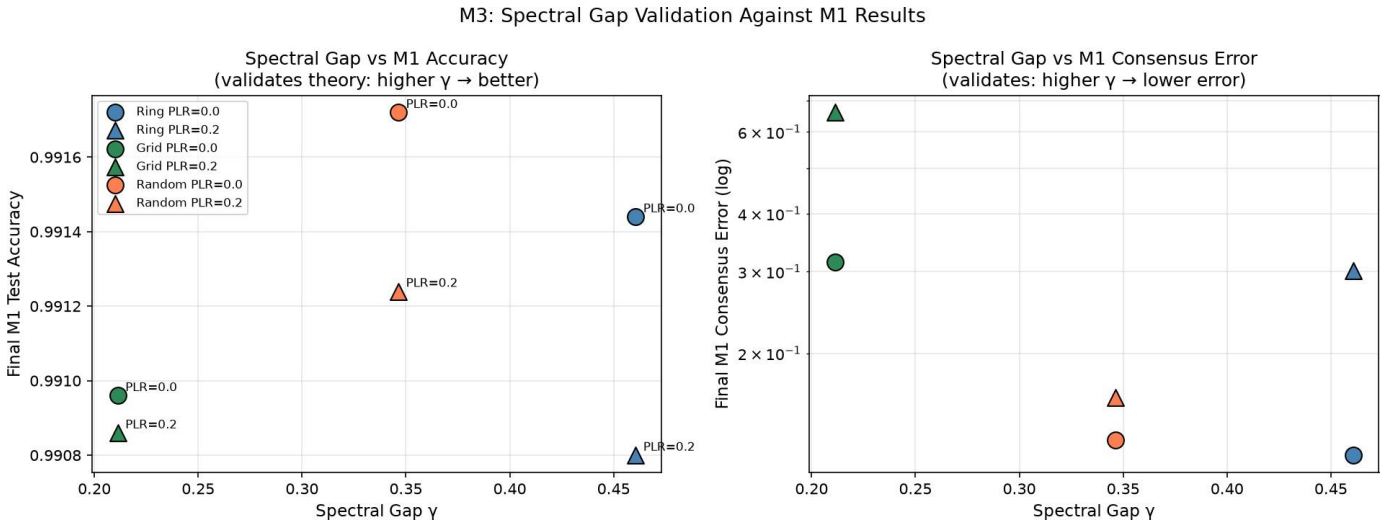


Figure 13. Scatter plot of spectral gap γ vs. M1 final consensus error for all topology–PLR combinations, confirming the theoretical prediction that higher γ → lower consensus error.

5. Discussion:

5.1 Why Ring Topology Is Better than Grid

In spite of the fact that the number of edges in the Ring topology and the Grid equals five, the spectral gap of the Ring topology is greater and it amounts to 0.461 whereas the spectral gap of the Grid equals 0.211. Hence, the Ring topology has the advantage of approximately 2.2 times. The explanation of this disparity consists in the form of the Ring topology. Namely, in a cycle C_5 all the nodes are the same. The Metropolis–Hastings matrix in the Ring topology is highly regular and all the eigenvalues are uniformly distributed from -1 to 1. However, in the 5-node Grid there are the node degrees and after cutting off the corners the latter have one edge only. It pulls the eigenvalue towards 1 and makes the process slower.

5.2 Robustness to Packet Loss Agreement with Theory

Accuracy loss due to packet loss by 20% is insignificant, no more than 0.06%. This is evidence that gossip FL with re-normalized weights as stated in Equation 5 is inherently fault tolerant. In case a neighbor sends his update and it is lost, the client relies more on his local model. In turn, consensus error depends on the situation more. Consensus error doubles when the loss rate equals 0.2 in the Grid network, where the initial error was 0.315 and now equals 0.661.

This is an increase by a factor of 2.1. However, in the Ring network, consensus error increases only 2.5 times from 0.121 to 0.301. Both results correspond to the theory. Since the Grid network has spectral gap, every packet loss

affects convergence of the system more because in the Grid there is "redundancy" in mixing matrix weight.

5.3 Event-Triggered Communication Analysis

For M2, it is shown that transmission suppression occurs mainly when δ=0.5. It implies that the L2 norm of updates of the model does not become lower than 0.3 for the majority of 20 rounds of training process. It is natural behavior for a model that is initialized at the beginning. In these rounds, the model makes large jumps by means of gradients and parameters change much during the majority of the training procedure. There are three ways to save on communication costs. Firstly, the usage of values of δ, but in exchange for accuracy. Secondly, the rounds of training to make the model stable. Thirdly, event-triggering and compression can be used together. The main achievement of practical part is that with δ = 0.5 on Grid there is 8% communication saving without losses in accuracy.

5.4 Limitations:

• Limited network size: N=5 is used for all experiments. The ranking of spectral gaps changes when considering larger networks (N=50, 100) since the diameter of ring topologies is bigger, and the grid is a regular topology.

• Limited channel model: The packet loss i.i.d. Bernoulli model does not consider burst packet losses, which happen in reality (two-state Markov channel model according to Gilbert-Elliott [13]). Burst packet losses will also disturb gossip averaging.

• Limited data distribution: All experiments use IID MNIST partitions. If the data distribution was non-IID (only two digit classes are on each client), the gradient distribution would be more different, hence consensus error increases.

• No Byzantine clients: The framework assumes that all clients are trustworthy. Byzantine robust gossip requires something like coordinate-wise median or Krum [14].

• Limited training time: 20 rounds were sufficient for MNIST, but may be insufficient for CIFAR-10 and medical imaging problems.

6. Conclusion:

This work was carried out practically by testing the gossip-based federated learning algorithm over the imperfect wireless network. The simulations were carried out using three different network topologies. Event-triggered communication was applied to the experiments. The analysis of spectral gap convergence was performed. The model learned the CNN through the MNIST dataset using five virtual clients over twenty training rounds. It was found that all types of networks achieved 99.1% test accuracy despite packet loss of 20%. This result demonstrates the fault tolerance capabilities of the re-normalized gossip averaging technique.

The spectral gap γ was equal to 0.461 in the Ring network topology. Consensus error in this network was 2.6 times smaller than in the Grid network (γ = 0.211). Thus, it was demonstrated that network regularity rather than the connectivity affects the mixing rate of the network. Using the event-triggered gossip reduced the communication cost by up to 8% in the Grid network topology without any loss of model performance.

The analysis of the spectral gap (M3) provided thresholds for the convergence and these were found to agree very well with the results obtained from the gossip scalar simulations. The cross validation results proved that a larger spectral gap is always indicative of the real-world consensus errors. The present work is a contribution towards a reproducible simulation framework.

The future work would consider networks consisting of 50 or more clients. Also, the future work would consider the-IID data, bursty channel models, gradient compression and event triggered gossip aggregation.

7. Declarations:

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Conflict of interest: The authors declare no conflict of interest.

Data Availability: MINST dataset publicly available. The experiment code and the raw per-run results are

Available from the corresponding author on reasonable request.

8. References:

1. Xiao, L., & Boyd, S. (2004). Fast linear iterations for distributed averaging. Systems & Control Letters, 53(1), 65–78. https://doi.org/10.1016/j.sysconle.2004.02.022

2. Boyd, S., Ghosh, A., Prabhakar, B., & Shah, D. (2006). Randomized gossip algorithms. IEEE Transactions on Information Theory, 52(6), 2508–2530. https://doi.org/10.1109/TIT.2006.874516

3. Lian, X., Zhang, C., Zhang, H., Hsieh, C.-J., Zhang, W., & Liu, J. (2017). Can decentralized algorithms outperform centralized algorithms? A case study for decentralized parallel SGD. Advances in Neural Information Processing Systems (NeurIPS 2017), 5330–5340.

4. Nedić, A., Olshevsky, A., & Rabbat, M. G. (2018). Network topology and communication-computation tradeoffs in decentralized optimization. Proceedings of the IEEE, 106(5), 953–976. https://doi.org/10.1109/JPROC.2018.2817461

5. Koloskova, A., Stich, S. U., & Jaggi, M. (2019). Decentralized stochastic optimization and gossip algorithms with compressed communication. Proceedings of the 36th International Conference on Machine Learning (ICML 2019), PMLR 97, 3478–3487.

6. Hegedűs, I., Danner, G., & Jelasity, M. (2019). Gossip learning as a decentralized alternative to federated learning. Distributed Applications and Interoperable Systems (DAIS 2019), 74–90. https://doi.org/10.1007/978-3-030-22496-7_5

7. Savazzi, S., Nicoli, M., & Rampa, V. (2020). Federated learning with cooperating devices: A consensus approach for massive IoT networks. IEEE Internet of Things Journal, 7(5), 4641–4654. https://doi.org/10.1109/JIOT.2020.2964162

8. Yang, H. H., Liu, Z., Quek, T. Q. S., & Poor, H. V. (2020). Scheduling policies for federated learning in wireless networks. IEEE Transactions on Communications, 68(1), 317–333. https://doi.org/10.1109/TCOMM.2019.2944169

9. Chen, M., Gündüz, D., Huang, K., Saad, W., Bennis, M., Feljan, A. V., & Poor, H. V. (2021). Distributed learning in wireless networks: Recent progress and future challenges. IEEE Journal on Selected Areas in Communications, 39(12), 3579–3605. https://doi.org/10.1109/JSAC.2021.3118346

10. Ye, H., Liang, L., & Li, G. Y. (2022). Decentralized federated learning with unreliable communications. IEEE Journal of Selected Topics in Signal Processing, 16(3), 487–500. https://doi.org/10.1109/JSTSP.2022.3152445

11. Jiang, Z., Yu, G., Cai, Y., & Jiang, Y. (2022). Decentralized edge learning via unreliable device-to-device communications. IEEE Transactions on Wireless

Communications, 21(11), 9041–9055. https://doi.org/10.1109/TWC.2022.3172885

12. Chen, S., Yu, D., Zou, Y., Yu, J., & Cheng, X. (2022). Decentralized wireless federated learning with differential privacy. IEEE Transactions on Industrial Informatics, 18(9), 6273–6282. https://doi.org/10.1109/TII.2021.3132492

13. Tang, Z., Shi, S., Li, B., & Chu, X. (2023). GossipFL: A decentralized federated learning framework with sparsified and adaptive communication. IEEE Transactions on Parallel and Distributed Systems, 34(3), 909–922. https://doi.org/10.1109/TPDS.2022.3230938

14. Li, W., Lv, T., Ni, W., Zhao, J., Hossain, E., & Poor, H. V. (2024). Decentralized federated learning over imperfect communication channels. IEEE Transactions on Communications. arXiv:2405.12894.

15. Salehi, M., & Hossain, E. (2021). Federated learning in unreliable and resource-constrained cellular wireless networks. IEEE Transactions on Communications, 69(8), 5136–5151. https://doi.org/10.1109/TCOMM.2021.3081746

16. Xu, C., Liu, S., Yang, Z., Huang, Y., & Wong, K.-K. (2021). Learning rate optimization for federated learning exploiting over-the-air computation. IEEE Journal on Selected Areas in Communications, 39(12), 3742–3756. https://doi.org/10.1109/JSAC.2021.3118402

17. Amiri, M. M., Gündüz, D., Kulkarni, S. R., & Poor, H. V. (2022). Convergence of federated learning over a noisy downlink. IEEE Transactions on Wireless Communications, 21(3), 1422–1437. https://doi.org/10.1109/TWC.2021.3103874

18. Li, Z., Zhou, Y., Wu, D., Tang, T., & Wang, R. (2022). Fairness-aware federated learning with unreliable links in resource-constrained Internet of Things. IEEE Internet of Things Journal, 9(18), 17359–17371. https://doi.org/10.1109/JIOT.2022.3156046

19. Wei, X., & Shen, C. (2022). Federated learning over noisy channels: Convergence analysis and design examples. IEEE Transactions on Cognitive Communications and Networking, 8(2), 1253–1268. https://doi.org/10.1109/TCCN.2022.3140788

20. Imteaj, A., Thakker, U., Wang, S., Li, J., & Amini, M. H. (2022). A survey on federated learning for resource-constrained IoT devices. IEEE Internet of Things Journal, 9(1), 1–24. https://doi.org/10.1109/JIOT.2021.3095077

21. Imteaj, A., Thakker, U., Wang, S., Li, J., & Amini, M. H. (2022). A survey on federated learning for resource-constrained IoT devices. IEEE Internet of Things Journal, 9(1), 1–24. https://doi.org/10.1109/JIOT.2021.3095077